PROJECT AZ – 900: TASK 7, IMPLEMENTING AN AZURE IoT Hub

- Task 1: Created an IoT Hub
- Task 2: Added an IoT Device
- Task 3: Tested the device using the Raspberry Pi Simulator or Central IoT Application (As a matter of fact, I explored all the recommended options while building or working on this project"

  I Downloaded IoT plug and play app (Play store or App store) to connected network device to IoT for management.

  NOTE: Auto Pi– Is a platform designed to connect network devices or a car to the cloud. It collects data of any device and send it to the cloud.

- For a car (Speed, Engine health, fuel efficiency …and more)

PROOF ON SCREENSHOTS

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Home > IoT Hub

IoT hub

Microsoft

Basics | Networking | Management | Add-ons | Tags | Review + create

Create an IoT hub to help you connect, monitor, and manage billions of your IoT assets. Learn more

Project details

Choose the subscription you'll use to manage deployments and costs. Use resource groups like folders to help you organize and manage resources.

Subscription * : Azure subscription 1

Resource group * : (New) bbiot-devices

Create new

Instance details

IoT hub name * : bbiotupon026

Region * : East US

Tier * : Standard (most popular)

Compare tiers

Daily message limit * : 400,000 ($25/month)

See all options

Review + create | < Previous | Next: Networking >

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IoT hub

Microsoft

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You can connect to your IoT hub either publicly via its public hostname or privately using a private endpoint.
Learn more

Connectivity configuration *

- (selected) Public access
- Private access (Recommended)

You can change this or configure another connectivity method after this resource has been created.
Learn more

Minimum TLS Version

- (selected) 1.2
- 1.0

TLS 1.0 and 1.1 are considered legacy and are planned for deprecation, but will be supported until further notice.
Learn more

Review + create | < Previous: Basics | Next: Management >

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IoT hub

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Role-based access control

Change the permission model to Azure role-based access control (RBAC) only, or to a combination of shared access policies and RBAC. Learn more

Permission model

- RBAC only
- (selected) Shared access policy + RBAC

To manage the elements within an instance, a user needs access to IoT Hub data APIs. Select the suggested role below to grant yourself full access to the APIs. You can also use Access Control (IAM) to choose appropriate roles later.
Learn more

Assign me: [ ] IoT Hub Data Contributor role

Scale

Device-to-cloud partitions * : 2

Review + create | < Previous: Networking | Next: Add-ons >

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Home > IoT Hub

IoT hub

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Tags are name/value pairs. To categorize resources and consolidate billing, apply the same tag to multiple resources and resource groups. Your tags will update automatically if you change your resources. Learn more

Name	Value	Resource
Creator	HilaryN	2 selected
		2 selected

Review + create | < Previous: Add-ons | Next: Review + create >

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Home

bbiotupon026-5284648 | Overview

Deployment

Search

- Overview
- Inputs
- Outputs
- Template

Delete | Cancel | Redeploy | Download | Refresh

Your deployment is complete

Deployment name : bbiotupon026-5284648
Subscription : Azure subscription 1
Resource group : bbiot-devices

Start time : 5/2/2026, 8:47:04 AM
Correlation ID : 770a345a-5ffb-4c94-b993-d7a1798ff310

Deployment details

Resource	Type	Status	Operation details
bbiotupon026	IoT Hub	OK	Operation details

Next steps

Go to resource

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >

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Home > bbiotupon026-5284648 | Overview

bbiotupon026

IoT Hub

Show me device metrics for this IoT Hub. | Summarize the properties of this IoT Hub. | How do I troubleshoot issues with this IoT Hub?

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Events
- Resource visualizer
- Device management
- Hub settings
- Security settings
- Defender for IoT
- Monitoring
- Automation
- Help

Move | Delete | Refresh | Feedback

See more

Usage | Get started

Show data for last: 1 Hour | 6 Hours | 12 Hours | 1 Day | 7 Days | 30 Days

IoT Hub Usage

- Messages used today: 0
- Daily messages quota: 400000
- IoT Devices: 0

Number of messages used

100 90 80 70 60 50 40 30 20 10 0

12 PM | 6 PM | May 2 | UTC+01:00

Total number of messages used (Max), bbio... 0

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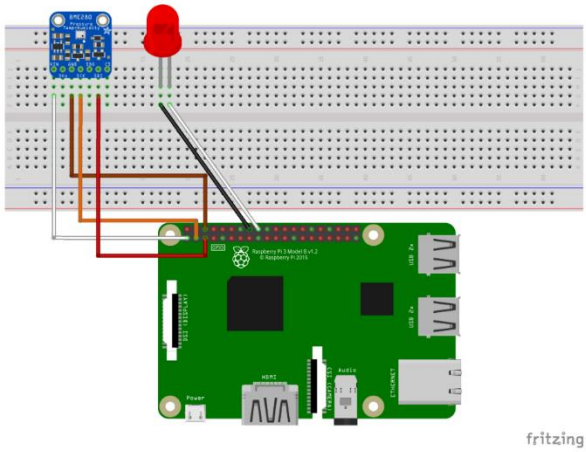
HilaryNwaja (Hilary Nwaja) | Raspberry Pi Azure IoT Web Sim

azure-samples.github.io/raspberry-pi-web-simulator/

Raspberry Pi Azure IoT Online Simulator

Help | English




```
/*
* IoT Hub Raspberry Pi NodeJS - Microsoft Sample Code - Copyright (c) 2017 - Licensed MIT
*/
const wpi = require('wiring-pi');
const Client = require('azure-iot-device').Client;
const Message = require('azure-iot-device').Message;
const Protocol = require('azure-iot-device-mqtt').Mqtt;
const BME280 = require('bme280-sensor');

const BME280_OPTION = {
  i2cBusNo: 1, // defaults to 1
  i2cAddress: BME280.BME280_DEFAULT_I2C_ADDRESS() // defaults to 0x77
};

const connectionString = '[Your IoT hub device connection string]';
const LEDPin = 4;

var sendingMessage = false;
var messageId = 0;
var client, sensor;
var blinkLEDTimeout = null;

function getMessage(cb) {
  messageId++;
  sensor.readSensorData()
    .then(function (data) {
      cb(JSON.stringify({
        messageId: messageId,
```

Run | Reset

Click 'Run' button to run the sample code(When sample is running, code is read-only).
Click 'Stop' button to stop the sample code running.
Click 'Reset' to reset the code.We keep your changes to the editor even you refresh the page.

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bbiotupon026 | Devices

IoT Hub

- Search
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- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Events
- Resource visualizer
- Device management
  - Devices
  - IoT Edge
  - Configurations + Deployments
  - Updates
  - Queries
- Hub settings
- Security settings
- Defender for IoT

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View, create, delete, and update devices in your IoT Hub. Learn more

Add Device | Edit columns | Refresh | Assign tags | Delete | Find devices using a query

enter device ID | Types: All | Add filter

Device ID	Type	Status	Last status update	Authentication type	C2D messages queued	Tags
2hokeafrix@	IoT Device	Enabled	--	Shared Access Signature	0	

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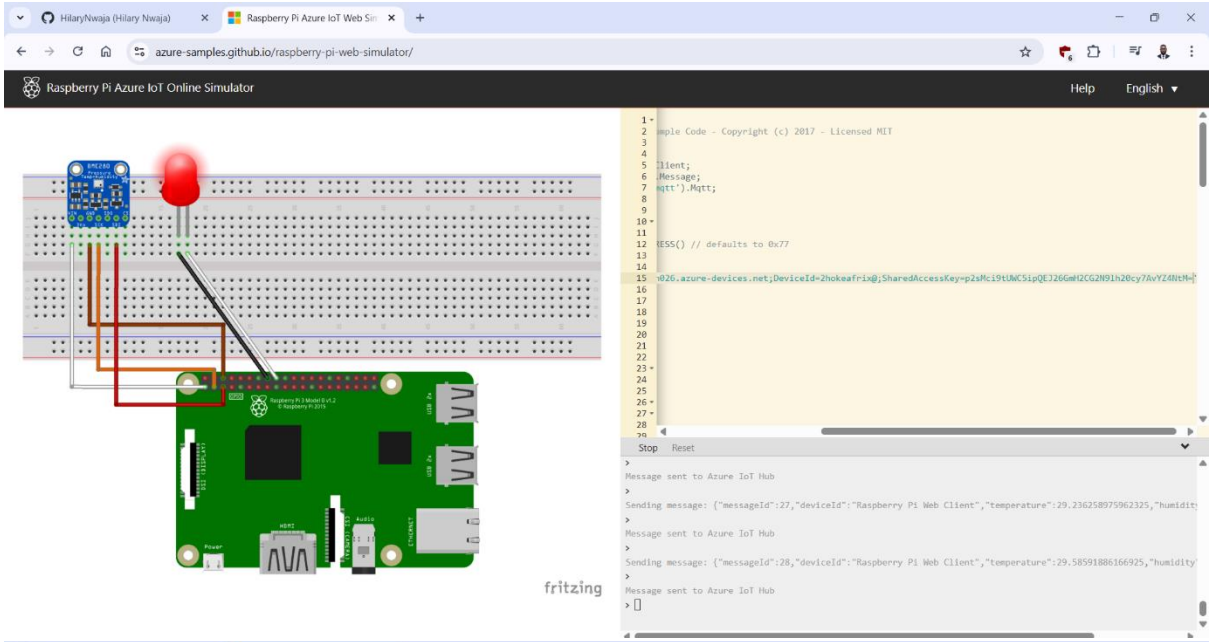
Home > bbiotupon026-5284648 | Overview > bbiotupon026 | Devices

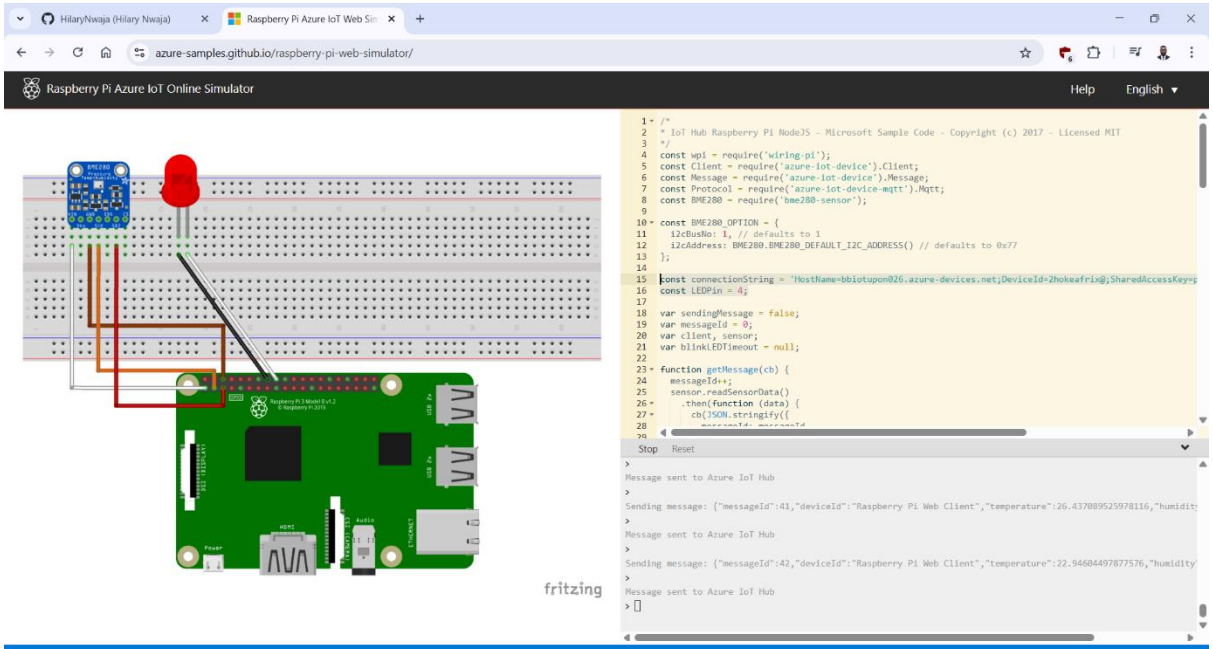
2hokeafrix@

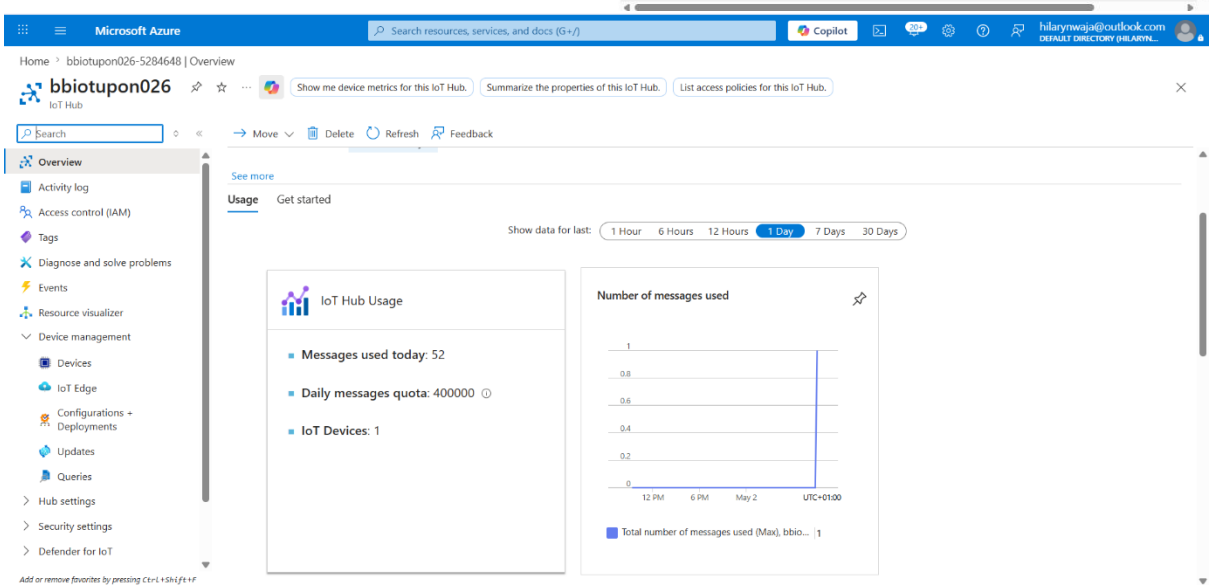
bbiotupon026

Save | Manage keys | Message to Device | Direct method | Add Module Identity | Device twin | Refresh

Device ID	2hokeafrix@
Primary key	••••••


azure-samples.github.io/raspberry-pi-web-simulator/
Raspberry Pi Azure IoT Online Simulator
Help
English
fritzing
Stop
Reset
Message sent to Azure IoT Hub
Sending message: {"messageId":27,"deviceId":"Raspberry Pi Web Client","temperature":29.236258975962325,"humidity
Message sent to Azure IoT Hub
Sending message: {"messageId":28,"deviceId":"Raspberry Pi Web Client","temperature":29.58591886166925,"humidity
Message sent to Azure IoT Hub


azure-samples.github.io/raspberry-pi-web-simulator/
Raspberry Pi Azure IoT Online Simulator
Help
English
fritzing
Stop
Reset
Message sent to Azure IoT Hub
Sending message: {"messageId":41,"deviceId":"Raspberry Pi Web Client","temperature":26.437089525978116,"humidity
Message sent to Azure IoT Hub
Sending message: {"messageId":42,"deviceId":"Raspberry Pi Web Client","temperature":22.94604497877576,"humidity
Message sent to Azure IoT Hub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bbiotupon026
IoT Hub
Show me device metrics for this IoT Hub.
Summarize the properties of this IoT Hub.
List access policies for this IoT Hub.
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Configurations + Deployments
Updates
Queries
Hub settings
Security settings
Defender for IoT
Add or remove favorites by pressing Ctrl+Shift+F
See more
Usage
Get started
Show data for last:
1 Hour
6 Hours
12 Hours
1 Day
7 Days
30 Days
IoT Hub Usage
Messages used today: 52
Daily messages quota: 400000
IoT Devices: 1
Number of messages used
1
0.8
0.6
0.4
0.2
0
12 PM
6 PM
May 2
UTC+01:00
Total number of messages used (Max), bbio... 1

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Home > IoT Central Applications

IoT Central Application

Create an IoT Central application with an application template. IoT Central is an IoT app platform that allows you to rapidly build enterprise-grade IoT solutions on a secure, reliable and scalable infrastructure. Learn more

Project details

Select the subscription to manage the deployed IoT Central resource and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : Azure subscription 1

Resource group * : bbiot-devices

Create new

Instance details

Resource name * : mybbiot-devices

Application URL * : mybbiot-devices.azureiotcentral.com

Template * : Smart Inventory Management

Region * : East US

Pricing plan * : Standard 2

Previous | Next | Review + create | Give feedback

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Home > IoT Central Applications

IoT Central Application

Basics | Tags | Review + create

View automation template

Basics

Subscription	Azure subscription 1
Resource group	bbiot-devices
Resource name	mybbiot-devices
Application URL	mybbiot-devices
Template	Smart Inventory Management
Region	East US
Pricing plan	Standard 2

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Home

CreateBlade-20260502092316 | Overview

Deployment

Search

- Overview
- Inputs
- Outputs
- Template

Delete | Cancel | Redeploy | Download | Refresh

Your deployment is complete

Deployment name : CreateBlade-20260502092316
Subscription : Azure subscription 1
Resource group : bbiot-devices

Start time : 5/2/2026, 9:27:21 AM
Correlation ID : 66d67396-fa65-41f5-b251-63238522af64

Deployment details

Resource	Type	Status	Operation details
mybbiot-devices	IoT Central Application	OK	Operation details

Next steps

Go to resource

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Microsoft Azure

Home > CreateBlade-20260502092316 | Overview

bbiot-devices
Resource group

How to manage these changes more efficiently with deployment tools? | Export resource groups using Bicep or Terraform | Generate Bicep code to duplicate this resource group.

Create | Manage view | Delete resource group | Refresh | Export to CSV | Open query | Assign tags | Move | Delete | Group by none

Essentials | JSON View

Resources | Recommendations

Filter for any field... | Type equals all | Location equals all | Add filter

Name	Type	Location
bbiotupon026	IoT Hub	East US
mybbiot-devices	IoT Central Application	East US

Showing 1 - 2 of 2. Display count: auto

Give feedback

mybbiot-devices.azureiotcentral.com/devices

mybbiot-devices | Search for devices

Devices | Filter templates | All devices | Inventory Gateways

Create a new device

To create a new device, select a device template, a name, and a unique ID. Learn more

Device name * : HlzIOT-SmartDevices

Device ID * : 2Hbb026GBM

Organization * : mybbiot-devices

Device template * : Unassigned

Simulate this device?
A simulated device generates telemetry that enables you to test the behavior of your application before you connect a real device.
No

Azure IoT Edge device?
Azure IoT Edge moves cloud analytics and custom business logic from the cloud to your devices.
No

Create | Cancel

mybbiot-devices.azureiotcentral.com/devices/details/2Hbb026GBM/rawdata

mybbiot-devices | Connect | Manage template

Devices > HlzIOT-SmartDevices

HlzIOT-SmartDevices
Last data received: N/A | Status: Registered

Raw data | Mapped aliases | Files

Timestamp	Message type

Device connection groups

ID scope : 0ne011BE1AF

Device ID : 2Hbb026GBM

Choose the connection type for this device. You can change this later if you need to.

Authentication type : Shared access signature (SAS)

Key | QR code

Shared Access Signatures (SAS) use security tokens and keys to connect to IoT Central. Use the SAS keys from the default enrollment group shown below to register your device. Learn more

Primary key : xn2Rhx394qKoWhk6BpVknGGdY7Ih32FPzGNAgzmbXAQ=

Secondary key : 6dYkmNUOv/MZHBavRmxVnalo1vUHrDFczx3yYkFvlME=

Close

mybbiot-devices.azureiotcentral.com/devices

mybbiot-devices | Search for devices

Devices

Filter templates

All devices

Inventory Gateways

IoT Plug and Play mobile

New | Import

All devices

Device explorer helps you see all your devices. Detailed information like device raw data helps you troubleshoot. Learn more

Device name	Device ID	Device status	Device template	Organization
HlzIOT-SmartDevices	2Hbb026GBM	Provisioned	IoT Plug and Play mobile	mybbiot-devices
BLE Gateway	BLEGateway_2gi5u4ttl0y	Provisioned	Inventory Gateways	mybbiot-devices
RFID Gateway	1ohns07jhs0	Provisioned	Inventory Gateways	mybbiot-devices

mybbiot-devices.azureiotcentral.com/devices/details/2Hbb026GBM/dtmi:azureiot:PhoneAsADevice:solutionModel:a0vftbw3s;1

mybbiot-devices | Search for devices

Connect | Manage template | Manage device

Devices > IoT Plug and Play mobile > HlzIOT-SmartDevices

HlzIOT-SmartDevices

Connected | Last data received: 5/2/2026, 9:47:19 AM | Status: Provisioned | Organization: mybbiot-devices

Device properties | Get started | Overview | Commands | Raw data | Mapped aliases | Files

Save

Device information

Device manufacturer: samsung

Device model: SM-S901U1

Operating system name: Android

Software version: 16

Total memory: 7571677184

Properties

Editable property

Cloud property

mybbiot-devices.azureiotcentral.com/devices/details/2Hbb026GBM/dtmi:azureiot:PhoneAsADevice:solutionModel:o343mkpcn;1

mybbiot-devices | Search for devices

Connect | Manage template | Manage device

Devices > IoT Plug and Play mobile > HlzIOT-SmartDevices

HlzIOT-SmartDevices

Connected | Last data received: 5/2/2026, 9:47:19 AM | Status: Provisioned | Organization: mybbiot-devices

Device properties | Get started | Overview | Commands | Raw data | Mapped aliases | Files

IoT Plug and Play app

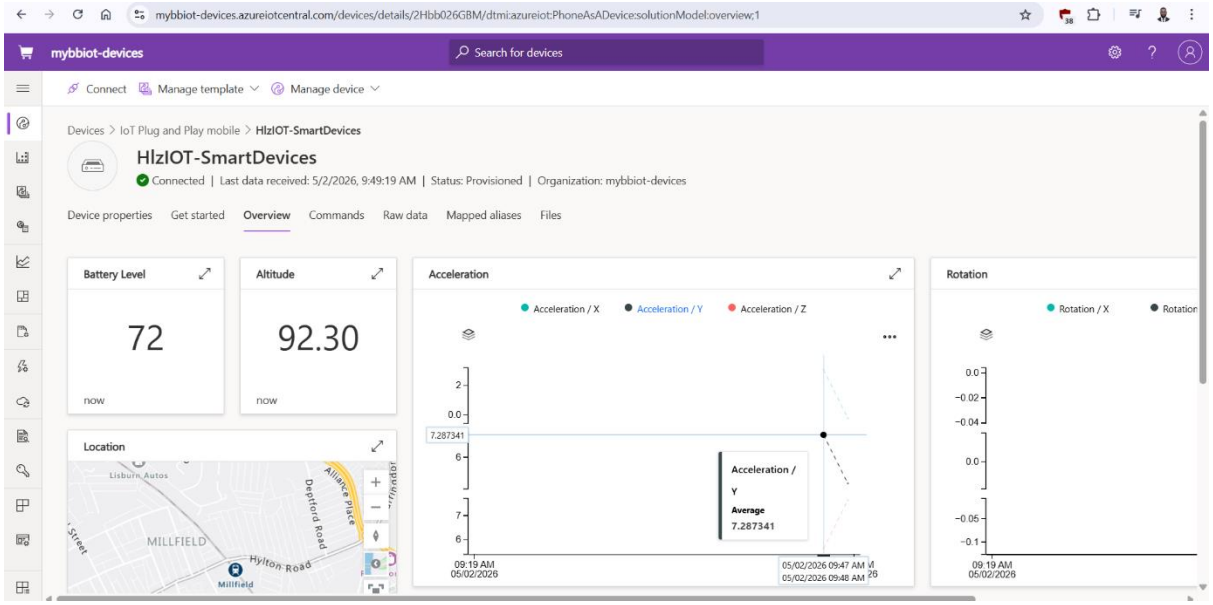
Turn your phone into an IoT device

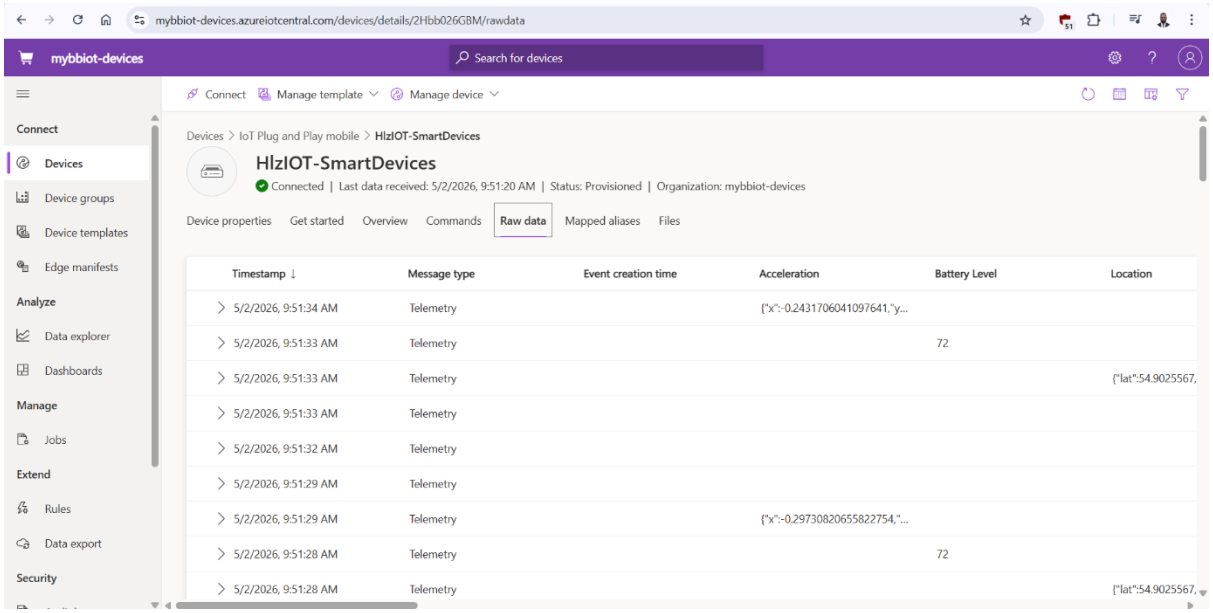
Download the IoT Plug and Play app on your mobile phone to get started

Follow the QR code or visit https://aka.ms/iot-paad-getstarted to learn more about the app.

Search

09:48
02/05/2026


mybbiot-devices.azureiotcentral.com/devices/details/2Hbb026GBM/dtmi:azureiot:PhoneAsADevice:solutionModel:overview;1
mybbiot-devices
Search for devices
Connect
Manage template
Manage device
Devices > IoT Plug and Play mobile > HlzIOT-SmartDevices
HlzIOT-SmartDevices
Connected | Last data received: 5/2/2026, 9:49:19 AM | Status: Provisioned | Organization: mybbiot-devices
Device properties
Get started
Overview
Commands
Raw data
Mapped aliases
Files
Battery Level
72
now
Altitude
92.30
now
Acceleration
Acceleration / X
Acceleration / Y
Acceleration / Z
7.287341
Acceleration / Y
Average
7.287341
09:19 AM
05/02/2026
Rotation
Rotation / X
Location
MILLFIELD
Hylton Road
Deptford Road
Alliance Place
Millfield


mybbiot-devices.azureiotcentral.com/devices/details/2Hbb026GBM/rawdata
mybbiot-devices
Search for devices
Connect
Manage template
Manage device
Devices > IoT Plug and Play mobile > HlzIOT-SmartDevices
HlzIOT-SmartDevices
Connected | Last data received: 5/2/2026, 9:51:20 AM | Status: Provisioned | Organization: mybbiot-devices
Device properties
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Overview
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Raw data
Mapped aliases
Files
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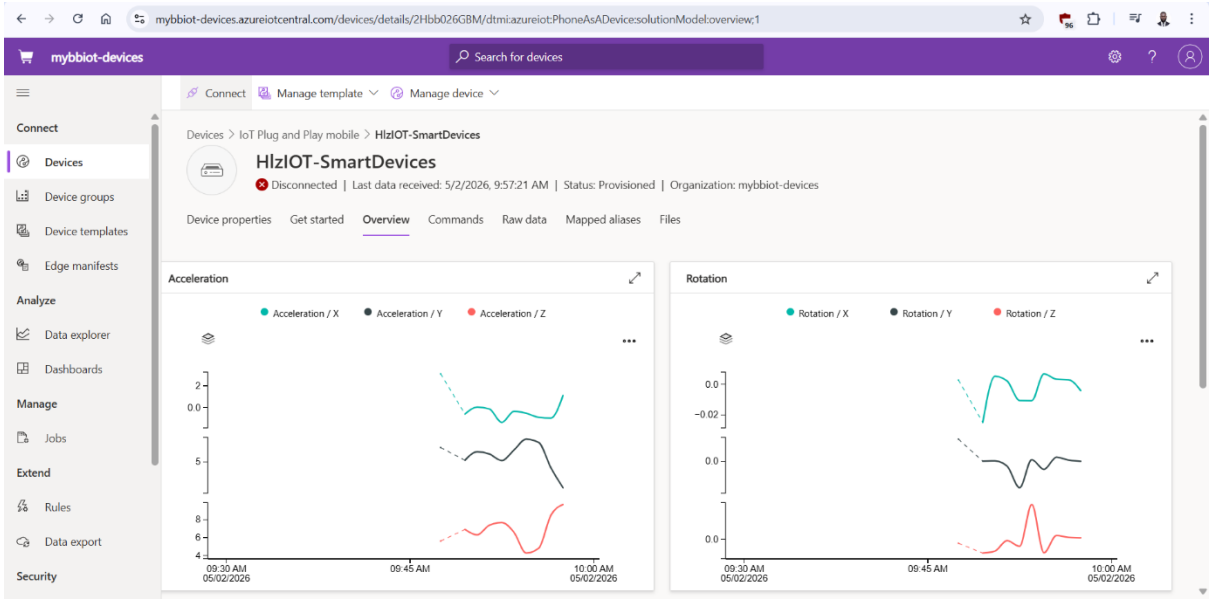
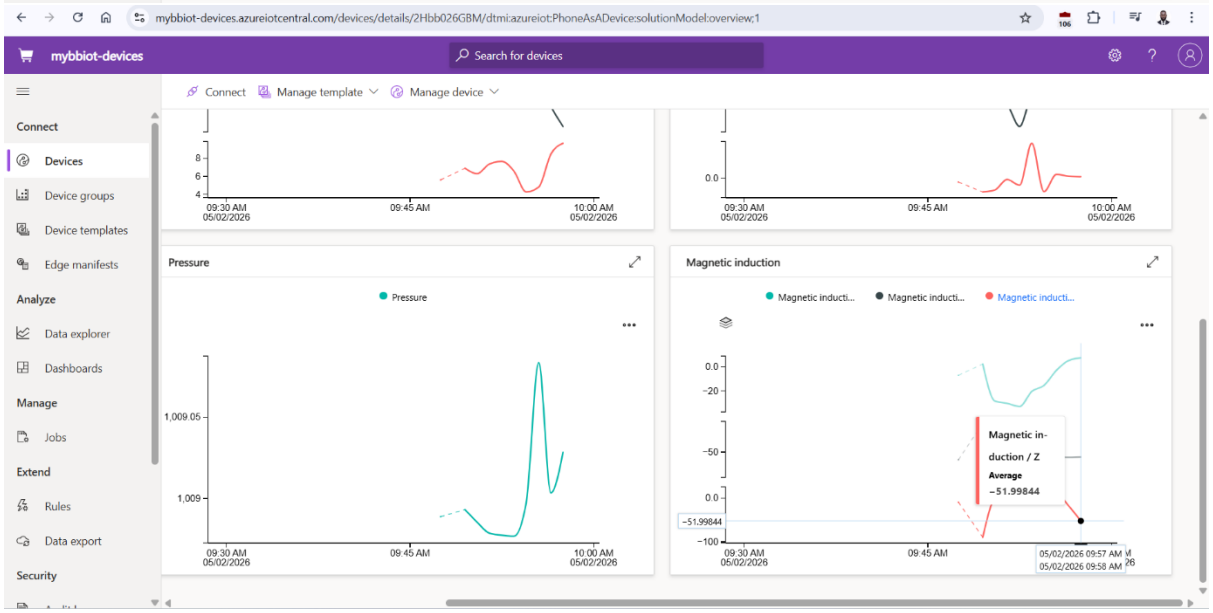
Device groups
Device templates
Edge manifests
Analyze
Data explorer
Dashboards
Manage
Jobs
Extend
Rules
Data export
Security
Timestamp
Message type
Event creation time
Acceleration
Battery Level
Location
5/2/2026, 9:51:34 AM Telemetry {"x":-0.2431706041097641,"y...
5/2/2026, 9:51:33 AM Telemetry 72
5/2/2026, 9:51:33 AM Telemetry {"lat":54.9025567,
5/2/2026, 9:51:33 AM Telemetry
5/2/2026, 9:51:32 AM Telemetry
5/2/2026, 9:51:29 AM Telemetry
5/2/2026, 9:51:29 AM Telemetry {"x":-0.2973082065582 2754,"...
5/2/2026, 9:51:28 AM Telemetry 72
5/2/2026, 9:51:28 AM Telemetry {"lat":54.9025567,


mybbiot-devices.azureiotcentral.com/devices/details/2Hbb026GBM/dtmi:azureiot:PhoneAsADevice:solutionModel:overview;1
mybbiot-devices
Search for devices
Connect
Manage template
Manage device
Connect
Devices
Device groups
Device templates
Edge manifests
Analyze
Data explorer
Dashboards
Manage
Jobs
Extend
Rules
Data export
Security
Devices > IoT Plug and Play mobile > HlzIOT-SmartDevices
HlzIOT-SmartDevices
Disconnected | Last data received: 5/2/2026, 9:57:21 AM | Status: Provisioned | Organization: mybbiot-devices
Device properties
Get started
Overview
Commands
Raw data
Mapped aliases
Files
Acceleration
Acceleration / X
Acceleration / Y
Acceleration / Z
Rotation
Rotation / X
Rotation / Y
Rotation / Z
09:30 AM 05/02/2026
09:45 AM
10:00 AM 05/02/2026


mybbiot-devices.azureiotcentral.com/devices/details/2Hbb026GBM/dtmi:azureiot:PhoneAsADevice:solutionModel:overview;1
mybbiot-devices
Search for devices
Connect
Manage template
Manage device
Connect
Devices
Device groups
Device templates
Edge manifests
Analyze
Data explorer
Dashboards
Manage
Jobs
Extend
Rules
Data export
Security
Pressure
Pressure
1,009.05
1,009
Magnetic induction
Magnetic inducti...
Magnetic in-
duction / Z
Average
−51.99844
−51.99844
09:30 AM 05/02/2026
09:45 AM
05/02/2026 09:57 AM
05/02/2026 09:58 AM